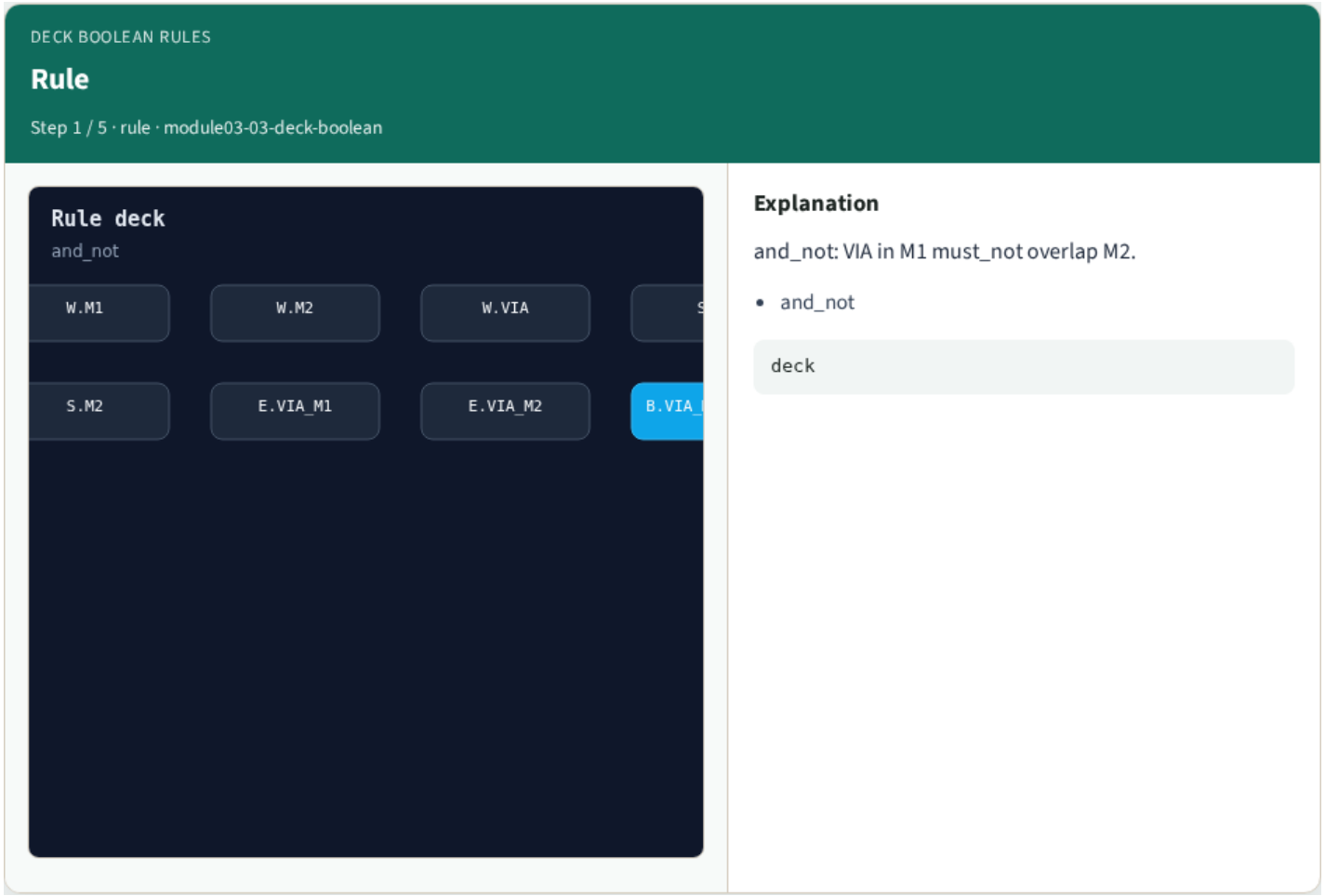
Boolean

Boolean deck rules express layer interactions , VIA covered by M1 and not M2

The idea

- GOLDEN: boolean hit=1 on V

Rule


DECK BOOLEAN RULES
Rule
Step 1 / 5 · rule · module03-03-deck-boolean
Rule deck
and_not
W.M1
W.M2
W.VIA
S.M2
E.VIA_M1
E.VIA_M2
Explanation
and_not: VIA in M1 must_not overlap M2.
and_not
deck

Rule

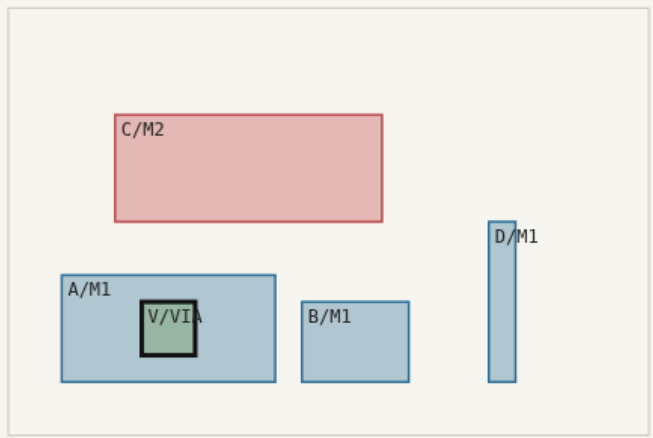
VIA

DECK BOOLEAN RULES

VIA

Step 2 / 5 · via · module03-03-deck-boolean

inner V



Explanation

Inner shape V on VIA layer.

- VIA

```
V
```

VIA

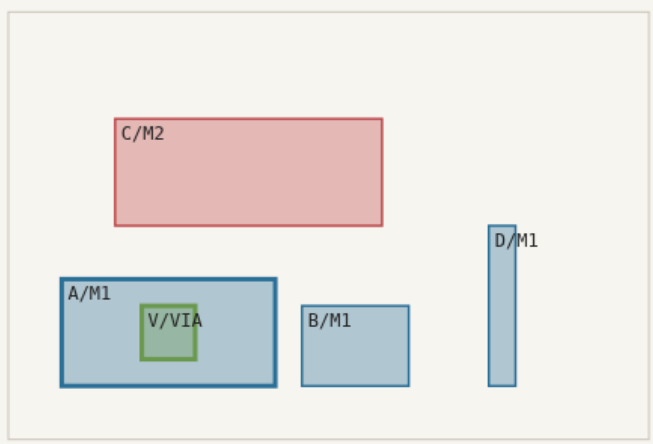
M1

DECK BOOLEAN RULES

M1

Step 3 / 5 · m1 · module03-03-deck-boolean

V∩A



Explanation

V overlaps M1 shape A.

- M1

```
A
```

M1

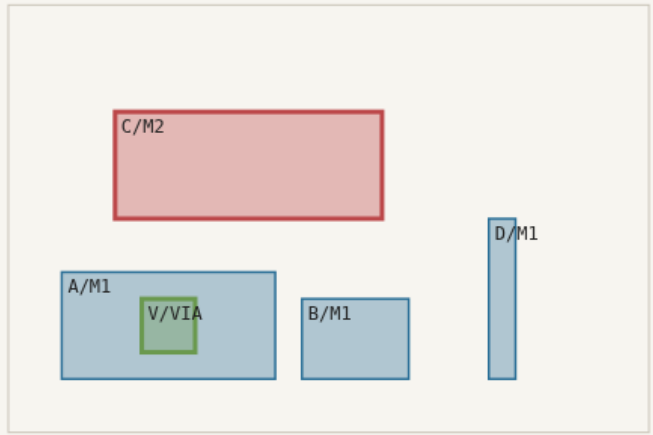
M2

DECK BOOLEAN RULES

M2

Step 4 / 5 · m2 · module03-03-deck-boolean

V∩C=false



Explanation

V does not overlap M2 shape C.

- M2

```
¬C
```

M2

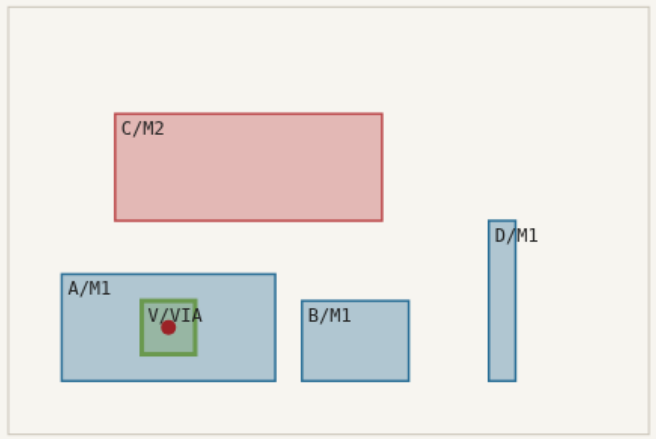
Golden

DECK BOOLEAN RULES

Golden

Step 5 / 5 · golden · module03-03-deck-boolean

boolean hit V



Explanation

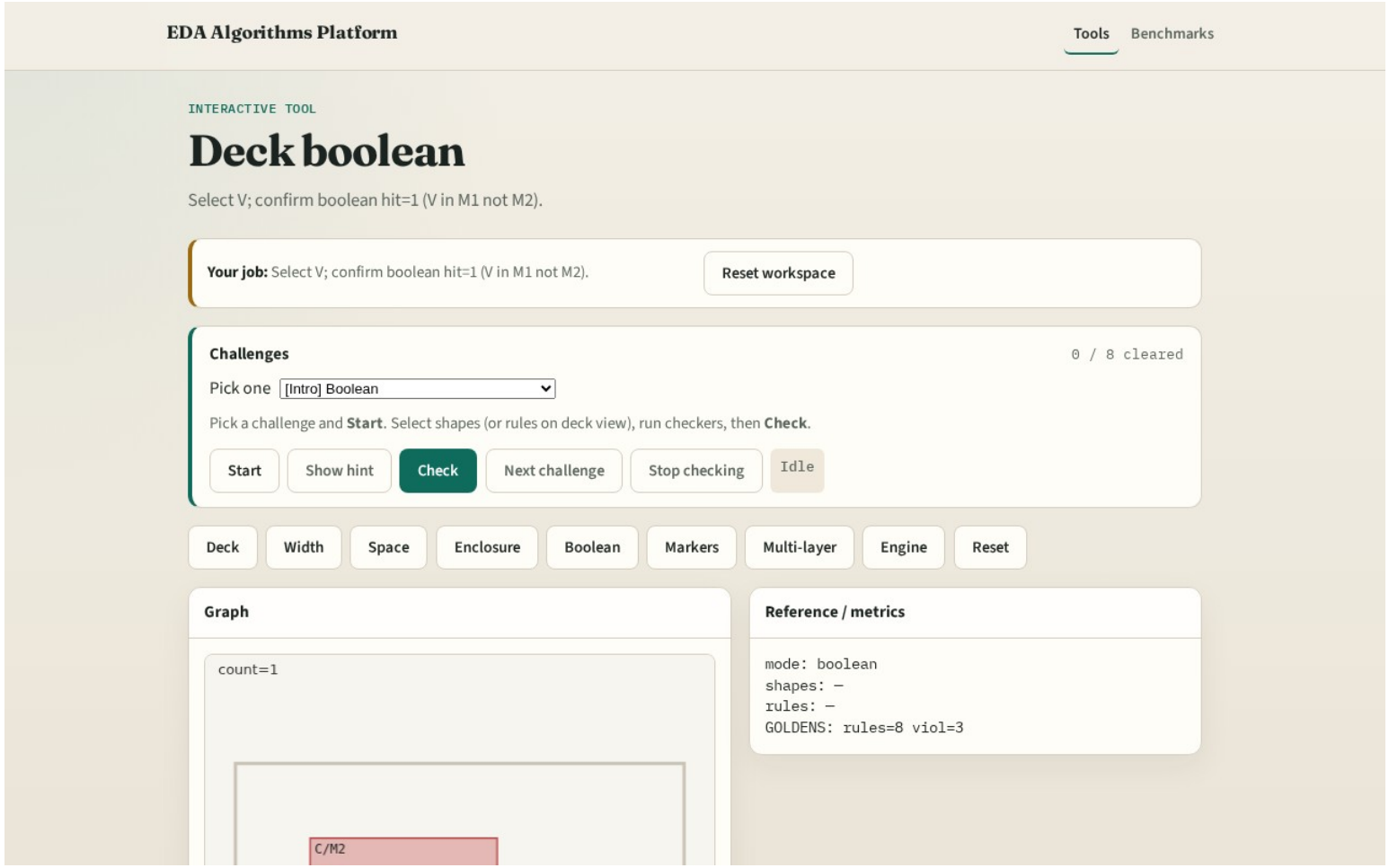
GOLDEN: boolean hit=1 on V.

- GOLDEN

```
1
```

Golden

Browser lab track



Browser lab starter

Implement track

- Call check_boolean(); assert count==1 and hits[0].shapes==['V']

Pitfalls

- Treating enclosure failures as boolean hits
- Requiring full containment for overlap

Your turn

- Next: DRC engine integration